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### SEMICONDUCTOR DEVICE

#### Abstract

A semiconductor device may include a first insulating interlayer on a substrate, first contact plugs passing through the first insulating interlayer, a second insulating interlayer on the first insulating interlayer and first contact plugs, and first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each first contact plug may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern, further covering a bottom of the first metal pattern, and further exposing an upper sidewall of the first metal pattern. Each of the first conductive patterns may have a first bottom surface contacting a portion of an upper surface of a respective first contact plug, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 USC § 119(a) to Korean Patent Application No. 10-2024-0033144, filed on Mar. 8, 2024, in the Korean Intellectual Property Office (KIPO), the contents of which are incorporated by reference herein in their entirety.

### BACKGROUND

[0002] Various example embodiments relate to a semiconductor device. Particularly, various example embodiments relate to a semiconductor device having high reliability.

[0003] Interconnection structures for electrically connecting circuit patterns to each other may be included in a semiconductor device. An interconnection structure may include a contact plug and a conductive pattern on the contact plug. As physical dimensions of the contact plug and the conductive pattern are becoming smaller, less space is present between the contact plug and conductive patterns adjacent to the contact plug in a horizontal direction. Accordingly, insulation breakdown of insulation material between the contact plug and the conductive pattern may occur, so that a reliability of the semiconductor device may be decreased.

### SUMMARY

[0004] Various example embodiments provide a semiconductor device including an interconnection structure having high reliability.

[0005] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a first insulating interlayer on the substrate, a plurality of first contact plugs passing through the first insulating interlayer, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern. Each of the plurality of first conductive patterns may have a first bottom surface contacting a portion of an upper surface of a respective first contact plug of the plurality of first contact plugs, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

[0006] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a first insulating interlayer on the substrate, a plurality of first contact plugs, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. The first insulating interlayer may include a plurality of contact holes. The plurality of first contact plugs may be in the plurality of contact holes, respectively. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern. An uppermost surface of the first barrier metal pattern may be lower than an upper surface of the first metal pattern. Each of the plurality of first conductive patterns may contact an upper surface of a respective first contact plug of the plurality of first contact plugs and may further contact a respective upper portion of the first insulating interlayer. A distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the

respective first contact plug and the first conductive pattern may be greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

[0007] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a plurality of capacitors on a cell region of the substrate, a plurality of peripheral circuit patterns on a peripheral circuit region of the substrate, a first insulating interlayer covering the plurality of capacitors and further covering the plurality of peripheral circuit patterns, a plurality of first contact plugs passing through the first insulating interlayer and connecting to the plurality of peripheral circuit patterns, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a first conductive pattern passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern. The first conductive pattern may contact upper surfaces of the plurality of first contact plugs and further contact an upper surface of the first insulating interlayer on the peripheral circuit region. A distance in a horizontal direction between an uppermost sidewall of the first metal pattern of a first contact plug of the plurality of first contact plugs and the first conductive pattern may be greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the first contact plug and the first conductive pattern.

[0008] According to example embodiments, in the interconnection structure, a minimum distance in a horizontal direction between the contact plug and the conductive pattern adjacent to the contact plug may be increased. Accordingly, insulation breakdown between the contact plug and the conductive pattern may be decreased, so that the interconnection structure may have high reliability.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Various example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings. FIGS. 1A to 35 represent various non-limiting, example embodiments as described herein.

[0010] FIG. 1A is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0011] FIG. 1B is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0012] FIG. 1C is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0013] FIG. 2 is a plan view of an interconnection structure of a semiconductor device according to example embodiments;

[0014] FIG. 3 is an enlarged cross-sectional view of a portion (i.e. portion C) of FIG. 1A;

[0015] FIG. 4 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments;

[0016] FIG. 5 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments;

[0017] FIGS. 6 to 15 are cross-sectional views illustrating a method of manufacturing an interconnection structure of a semiconductor device according to example embodiments;

[0018] FIG. 16 is a cross-sectional view of an interconnection structure of a semiconductor device

according to example embodiments;

[0019] FIG. **17** is an enlarged cross-sectional view of a portion of FIG. **16**;

[0020] FIG. **18** is a cross-sectional view of a semiconductor device according to example embodiments;

[0021] FIG. **19** is a cross-sectional view of a semiconductor device according to some example embodiments;

[0022] FIGS. **20** to **34** are cross-sectional views illustrating a method of manufacturing a semiconductor device according to example embodiments; and

[0023] FIG. **35** is a cross-sectional view of a semiconductor device according to example embodiments.

## DETAILED DESCRIPTION

[0024] Hereinafter, various example embodiments will be described in detail with reference to the accompanying drawings.

[0025] FIG. **1A** is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. **1B** is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. **1C** is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. **2** is a plan view of an interconnection structure of a semiconductor device according to example embodiments. FIG. **3** is an enlarged cross-sectional view of a portion (i.e. portion C) of FIG. **1A**. FIG. **4** is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments. FIG. **5** is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments.

[0026] Each of FIGS. **1A**, **1B**, and **1C** is a cross-sectional view taken along line I-I' of FIG. **2**. The interconnection structures shown in FIGS. **4** and **5** may be substantially the same as the interconnection structure shown in FIG. **1A**, except for materials occupying a first recess or a shape of the materials occupying the first recess.

[0027] Referring to FIGS. **1A**, **1B**, **1C**, **2**, and **3**, a first insulating interlayer **102** may be on a substrate **100**. In example embodiments, the first insulating interlayer **102** may include a silicon oxide-based material. For example, the first insulating interlayer **102** may include Tetra Ethyl Ortho Silicate (TEOS).

[0028] First contact holes **104** may pass through the first insulating interlayer **102**, and may expose a surface of the substrate **100**. In example embodiments, the surface of the substrate **100** may be exposed by bottoms of the first contact holes **104**. In some example embodiments, a lower conductive structure may be further on the substrate **100**, and the lower conductive layer may be exposed by bottoms of the first contact holes **104**.

[0029] In example embodiments, in a plan view, each of the first contact holes **104** may have a circular shape or an oval shape.

[0030] A first contact plug **112a** may be included for each of the first contact holes **104**, each first contact plug **112b** being in a respective one of the first contact holes **104**. The first contact plugs **112a** may pass through the first insulating interlayer **102**. In example embodiments, bottoms of the first contact plugs **112a** may contact the surface of the substrate **100**. In some example embodiments in which a lower conductive structure is further on the substrate **100**, the bottoms of the first contact plugs **112a** may contact the lower conductive structure.

[0031] The first contact plugs **112a** may each include a first barrier metal pattern **106b** and a first metal pattern **110a**. The first barrier metal pattern **106b** may cover a portion of a sidewall of the first metal pattern **110a** and further cover a bottom of the first metal pattern **110a**.

[0032] The first barrier metal pattern **106b** may conform to the portion of the sidewall of the first metal pattern **110a** and further to the bottom of each of the first contact holes **104**. In example embodiments, the first barrier metal pattern **106b** on the portion of the sidewall of the first metal

pattern **110a** and on the bottom of each of the first contact holes **104** may have a uniform thickness. The first metal pattern **110a** may be on the first barrier metal pattern **106b** to occupy the first contact hole **104**. Accordingly, the first barrier metal pattern **106b** may surround the portion of the sidewall of the first metal pattern **110a** and may further surround the bottom of the first metal pattern **110a**.

[0033] An upper surface of the first metal pattern **110a** may be substantially coplanar with an upper surface of the first insulating interlayer **102**. An uppermost surface of the first barrier metal pattern **106b** may be lower than the upper surface of the first metal pattern **110a**. The first barrier metal pattern **106b** may not be on an upper sidewall of the first metal pattern **110a**. An upper sidewall of the first metal pattern **110a** may be exposed by the first barrier metal pattern **106b**. The first barrier metal pattern **106b** may not be disposed between the upper sidewall of the first metal pattern **110a** and the first insulating interlayer **102**.

[0034] A first recess **114** may be between the upper sidewall of the first metal pattern **110a** and the first insulating interlayer **102**, and the first recess **114** may surround the upper sidewall of the first metal pattern **110a**. The first recess **114** may be disposed above the uppermost surface of the first barrier metal pattern **106b**.

[0035] In example embodiments, a sidewall of the first contact hole **104** may have a vertical slope, or may have a sidewall slope such that an inner width of the first contact hole **104** decreases from upper portion to the bottom thereof. Accordingly, a sidewall of the first contact plug **112a** may have the vertical slope, or may have a sidewall slope such that a width of the first contact plug **112a** decreases from upper portion to the bottom thereof.

[0036] In example embodiments, the first barrier metal pattern **106b** may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, or tungsten nitride.

[0037] In example embodiments, the first metal pattern **110a** may include tungsten. In some example embodiments, the first metal pattern **110a** may include aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0038] A second insulating interlayer **120** may occupy the first recess **114**, and may cover upper surfaces of the first contact plugs **112a** and of the first insulating interlayer **102**. The second insulating interlayer **120** may include a silicon oxide-based material. The second insulating interlayer **120** may include a material having a dielectric constant lower than a dielectric constant of the first insulating interlayer **102**. The second insulating interlayer **120** may include a low-K insulation material. For example, the second insulating interlayer **120** may include SiOC or SiOF. The same insulation material (e.g., the second insulating interlayer) may be in the first recess **114**, on the first insulating interlayer **102**, and on the first contact plug **112a**. For example, the first recess **114** may not be occupied with nitride, such as silicon nitride.

[0039] In example embodiments, as shown in FIGS. **1A**, **1B**, **1C**, and **3**, voids may not be included in the second insulating interlayer **120** inside the first recess **114**. In some example embodiments, as shown in FIG. **4**, a void **118** may be included in the second insulating interlayer **120** inside the first recess **114**.

[0040] In some example embodiments, as shown in FIG. **5**, the first recess **114** may not be occupied with the second insulating interlayer **120** therein, so that the first recess **114** may remain as a space including, e.g., air **119**. In this case, the second insulating interlayer **120** may be disposed on the first insulating interlayer **102** and on the first contact plug **112a** to cover an inlet portion of the first recess **114**.

[0041] Openings **122** may pass through the second insulating interlayer **120**, and may extend below the upper surface of the first insulating interlayer **102** in a vertical direction (i.e., a direction perpendicular to the upper surface of the substrate **100**). The first contact plug **112a** may be exposed by at least a lower portion of each of the openings **122**. Each of the openings **122** may have a sidewall slope such that an inner width of each of the openings **122** decreases from an upper portion to a bottom thereof. Each of the openings **122** may have a lower width that is less than an

upper width.

[0042] A first conductive pattern **134** may be in each of the openings **122**. The first conductive pattern **134** may pass through the second insulating interlayer **120**, and may extend below the upper surface of the first insulating interlayer **102**.

[0043] A first bottom portion of the first conductive pattern **134** may contact the upper surface of the first contact plug **112a**. A second bottom portion of the first conductive pattern **134** may contact the first insulating interlayer **102**.

[0044] In example embodiments, the first conductive pattern **134** may not entirely cover the upper surface of a first contact plug **112a**, but may partially cover the upper surface of a first contact plug **112a**. In example embodiments, a lower portion of the first conductive pattern **134** may contact the upper surface of the first metal pattern **110a** and further contact an upper sidewall of the first metal pattern **110a**, included in the first contact plug **112a**, and the lower portion of the first conductive pattern **134** may further contact an upper surface of the first barrier metal pattern **106b**. In example embodiments, a portion of the first conductive pattern **134** may be disposed in a region corresponding to but opposite from the first recess **114** relative to, e.g., the first contact plug **112a**. The first bottom portion of the first conductive pattern **134** may be higher than the second bottom portion of the first conductive pattern **134**. In example embodiments, the first bottom portion of the first conductive pattern **134** may be substantially coplanar with the upper surface of the first metal pattern **110a** of the first contact plug **112a**. The second bottom portion of the first conductive pattern **134** may be lower than the upper surface of the first insulating interlayer **102**, and the second bottom portion further may be higher than a bottom of the first insulating interlayer **102**. The second bottom portion of the first conductive pattern **134** may be at an inner portion of the first insulating interlayer **102**.

[0045] In example embodiments, as shown in FIGS. **1A** and **1C**, the second bottom portion of the first conductive pattern **134** may be substantially coplanar with a bottom of the first recess **114**. The second bottom portion of the first conductive pattern **134** may be substantially coplanar with the uppermost surface of the first barrier metal pattern **106b**. In some example embodiments, as shown in FIG. **1B**, the second bottom portion of the first conductive pattern **134** may be lower than the bottom of the first recess **114**. The second bottom portion of the first conductive pattern **134** may be lower than the uppermost surface of the first barrier metal pattern **106b**.

[0046] The first conductive pattern **134** may include a second barrier metal pattern **130a** and a second metal pattern **132a**. The second barrier metal pattern **130a** may cover a sidewall of the second metal pattern **132a** and may further cover a bottom of the second metal pattern **132a**.

[0047] The second barrier metal pattern **130a** may conform to the sidewall of the opening **122** and to the bottom of the opening **122**. In example embodiments, the second barrier metal pattern **130a** on the sidewall and on the bottom of the opening **122** may have a uniform thickness. In some example embodiments, the second barrier metal pattern **130a** on the bottom of the opening **122** may have a thickness greater than a thickness of the second barrier metal pattern **130a** on the sidewall of the opening **122**.

[0048] The second metal pattern **132a** may be on the second barrier metal pattern **130a** to occupy the opening **122**.

[0049] Upper surfaces of the second metal pattern **132a** and of the second barrier metal pattern **130a** may be substantially coplanar with an upper surface of the second insulating interlayer **120**.

[0050] In example embodiments, the second barrier metal pattern **130a** may include titanium, titanium nitride, tantalum, tantalum nitride, tungsten nitride, or the like. The second metal pattern **132a** may include, e.g., copper. In some example embodiments, the second metal pattern **132a** may include aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, gold or the like.

[0051] In the first conductive pattern **134**, a portion lower than the upper surface of the first insulating interlayer **102** is referred to within the following disclosure as a first portion of the first conductive pattern **134**, and a portion higher than the upper surface of the first insulating interlayer

is referred to within the following disclosure as a second portion of the first conductive pattern **134**. In the claims, the recitation of a “first portion” or a “second portion” without more is not intended to incorporate these definitions therein. The first portion of the first conductive pattern **134** may contact the first insulating interlayer **102** without contacting the upper surface of the first contact plug **112a**.

[0052] A width of an uppermost portion of the first conductive pattern **134** may be greater than a width of a lowermost portion of the first conductive pattern **134**. At least the first portion of the first conductive pattern **134** may have a sidewall slope such that the width of first conductive pattern **134** gradually decreases downward.

[0053] In example embodiments, as shown in FIGS. **1A** and **1B**, the first conductive pattern **134** may have the sidewall slope such that the width of the first conductive pattern **134** gradually decreases from the uppermost portion of the first conductive pattern **134** to the lowermost portion of the first conductive pattern **134**. The sidewall slope of the first conductive pattern **134** may be constant. In some example embodiments, as shown in FIG. **1C**, an upper sidewall of the first conductive pattern **134** may have a vertical slope, and a lower sidewall of the first conductive pattern **134** may have a slope such that the width of first conductive pattern **134** gradually decreases downward.

[0054] In example embodiments, a portion of at least one first conductive pattern contacting the first insulating interlayer **102** may be disposed between portions of two first conductive patterns **134** contacting the upper surface of the first contact plug **112a**. The first portion of the first conductive pattern **134** may be disposed adjacent to the first contact plug **112a** while being spaced apart from the first contact plug **112a**. In a cross-sectional view, an upper sidewall of the first contact plug **112a** and the first portion of the first conductive pattern **134** may face toward each other in a horizontal direction with respect to the upper surface of the substrate.

[0055] In example embodiments, an upper portion of the first portion of the first conductive pattern **134** may face the sidewall of the first metal pattern **110a**, and a lower portion of the first portion of the first conductive pattern **134** may face a sidewall of the first barrier metal pattern **106b**. In some example embodiments, the first portion of the first conductive pattern **134** may face only the sidewall of the first metal pattern **110a**.

[0056] A distance **d2** in the horizontal direction between an uppermost sidewall of the first metal pattern **110a** and the first conductive pattern **134** may be greater than a distance **d1** in the horizontal direction between an uppermost sidewall of the first barrier metal pattern **106b** and the first conductive pattern **134**.

[0057] The first and second insulating interlayers **102** and **120** may be laterally disposed between the uppermost sidewall of the first contact plug **112a** and the first portion of the first conductive pattern **134**.

[0058] The second insulating interlayer **120** and the first insulating interlayer **102** may be disposed in the horizontal direction between the sidewall of the first metal pattern **110a** positioned higher than the uppermost surface of the first barrier metal pattern **106b** and the first portion of the first conductive pattern **134**. Only the first insulating interlayer **102** may be disposed between the sidewall of the first barrier metal pattern **106b** and the first portion of the first conductive pattern **134**.

[0059] In the following disclosure, the distance in the horizontal direction between the uppermost sidewall of the first barrier metal pattern **106b** and the first conductive pattern **134** is referred to as the first distance **d1**. A portion having the first distance is referred to in the following disclosure as a third portion. In the claims, the recitation of a “first distance” or a “third portion” without more is not intended to incorporate these definitions therein. A third portion may be a weak region where insulation breakdown of insulation material between the first contact plug **112a** and the first conductive pattern **134** easily occurs. Therefore, the weak region may compromise the reliability of the semiconductor device. In order to improve reliability of the semiconductor device, it is

desirable to increase the first distance **d1**. Additionally, when an interface between the insulation materials (e.g., an interface between the first and second insulation interlayers) is positioned at the third portion, the insulation breakdown of the insulation material may easily occur at the interface between the insulation materials. Therefore, preferably, the interface between the insulation materials may not be positioned at the third portion.

[0060] An uppermost portion of the first portion of the first conductive pattern may have a width greater than a width of a lowermost portion of the first portion of the first conductive pattern. Therefore, if an upper portion of the first barrier metal pattern **106b** is not recessed, the first distance **d1** may be greatly decreased. In addition, if the upper portion of the first barrier metal pattern is not recessed, an interface between the upper surface of the first insulating interlayer and a lower surface of the second insulating interlayer may be positioned at the third portion to have a plane shape. In this case, the third portion may correspond to the interface between the upper surface of the first insulating interlayer and the lower surface of the second insulating interlayer. The interface between the first and second insulating interlayers may be may coincide with the third portion, and the first distance may be decreased. Therefore, insulation breakdown of insulation material positioned at the third position may easily occur.

[0061] However, in example embodiments, the uppermost surface of the first barrier metal pattern **106b** may be lower than the upper surface of the first metal pattern **110a**, and thus the first recess **144** may be included in the first contact plug **112a**. In this case, the third portion may be positioned at a level lower than the uppermost surface of the first contact plug **112a**. The third portion may not correspond to a portion between an uppermost portion of the first contact plug **112a** and the first conductive pattern **134**.

[0062] In addition, the first distance **d1** in the case in which the upper portion of the first barrier metal pattern **106b** is recessed may be greater than the first distance in the case in which the upper portion of the first barrier metal pattern **106b** is not recessed.

[0063] The interface between the upper surface of the first insulating interlayer **102** and the lower surface of the second insulating interlayer **120** may not be positioned at the third portion. Only the first insulating interlayer **102** may be disposed at the third portion. Accordingly, the interface between the insulating interlayers positioned at the third portion may be more reliable.

[0064] The first distance **d1** may be increased, and the interface between the first and second insulating interlayers **102** and **120** may not be positioned at the third portion. Accordingly, insulation breakdown of insulation material positioned at the third portion may be less likely.

[0065] A capping layer **140** may be disposed on the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may cover the upper surfaces of the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0066] FIGS. **6** to **15** are cross-sectional views illustrating a method of manufacturing an interconnection structure of a semiconductor device according to example embodiments.

[0067] Referring to FIG. **6**, a first insulating interlayer **102** may be formed on the substrate **100**.

[0068] The substrate **100** may include silicon, germanium, silicon-germanium, or a group III-V compound such as GaP, GaAs, or GaSb. In some example embodiments, the substrate **100** may be a Silicon On Insulator (SOI) substrate or a Germanium On Insulator (GOI) substrate.

[0069] The first insulating interlayer **102** may include a silicon oxide-based material. For example, the first insulating interlayer **102** may include Tetra Ethyl Ortho Silicate (TEOS). In example embodiments, before forming the first insulating interlayer **102**, a lower conductive structure may further be formed on an upper surface of the substrate **100**.

[0070] An etch mask pattern may be formed on the first insulating interlayer **102**, and the first insulating interlayer **102** may be etched using the etch mask pattern to form first contact holes **104**. In example embodiments, the etch mask pattern may have a structure in which an amorphous



carbon layer pattern and a silicon oxynitride layer pattern are stacked. The etch mask pattern may be formed by deposition processes of the amorphous carbon layer and the silicon oxynitride layer and photolithography processes of the amorphous carbon layer and the silicon oxynitride layer. [0071] The etching process of the first insulating interlayer **102** may include a dry etching process. [0072] Referring to FIG. 7, a first barrier metal layer **106** may be conformally formed on sidewalls and bottoms of the first contact holes **104** and on an upper surface of the first insulating interlayer **102**.

[0073] In example embodiments, the first barrier metal layer **106** may be formed to have a uniform thickness on the sidewalls and on the bottoms of the first contact holes **104** and the upper surface of the first insulating interlayer **102**.

[0074] The first barrier metal layer **106** may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The first barrier metal layer **106** may be formed by a chemical vapor deposition process or an atomic layer deposition process.

[0075] Referring to FIG. 8, the first metal layer **110** may be formed on the first barrier metal layer **106** to fill the first contact holes **104**.

[0076] The first metal layer **110** may include, e.g., tungsten. In this case, the first metal layer **110** may be formed by a chemical vapor deposition process or an atomic layer deposition process using at least WF<sub>6</sub> as a deposition source.

[0077] In some example embodiments, the first metal layer **110** may include aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, gold, etc.

[0078] Referring to FIG. 9, the first barrier metal layer **106** and the first metal layer **110** may be planarized until the upper surface of the first insulating interlayer **102** is exposed to form a preliminary first contact plug **112** in each of the first contact holes **104**. The preliminary first contact plug **112** may include a preliminary first barrier metal pattern **106a** and a first metal pattern **110a**. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0079] Referring to FIG. 10, an upper portion of the preliminary first barrier metal pattern **106a** may be partially etched to form a first barrier metal pattern **106b**. Accordingly, the first contact plug **112a** including the first barrier metal pattern **106b** and the first metal pattern **110a** may be formed in each of the first contact holes **104**.

[0080] An uppermost surface of the first barrier metal pattern **106b** may be lower than an upper surface of the first metal pattern **110a**. A first recess **114** may be formed between the first insulating interlayer **102** and the first metal pattern **110a** above the uppermost surface of the first barrier metal pattern **106b**. In the first contact plug **112a**, an upper outer wall of the first metal pattern **110a** may be exposed by the first recess **114**.

[0081] Referring to FIG. 11, a second insulating interlayer **120** may be formed on the first insulating interlayer **102** and the first contact plug **112a** to fill the first recess **114**. The second insulating interlayer **120** may cover the first insulating interlayer **102** and the first contact plug **112a**. The same insulation material (e.g., the second insulating interlayer) may be formed in the first recess **114** and on the first insulating interlayer **102** and the first contact plug **112a**.

[0082] In example embodiments, voids may not be present in the second insulating interlayer **120** formed in the first recess **114**.

[0083] In some example embodiments, voids may be present in the second insulating interlayer **120** formed in the first recess **114**. In this case, the interconnection structure shown in FIG. 4 may be formed by subsequent processes.

[0084] In some example embodiments, the second insulating interlayer **120** may not fill the first recess **114**. The second insulating interlayer **120** may be formed on the first insulating interlayer **102** and the first contact plug **112a** to cover an inlet of the first recess **114**. Accordingly, the first recess **114** may be empty. In this case, the interconnection structure shown in FIG. 5 may be formed by subsequent processes.

[0085] Referring to FIGS. 12 and 13, portions of the second insulating interlayer 120 and the first insulating interlayer 102 may be etched to form openings 122. Processes for forming the openings 122 may include a photo process and an etching process. The etching process may include, e.g., a dry etching process.

[0086] The openings 122 may pass through the second insulating interlayer 120, and may extend into a portion under the upper surface of the first insulating interlayer 102 in the vertical direction. A first portion of a bottom of each of the openings 122 may expose an upper surface of the first contact plug 112a. A second portion of the bottom of each of the openings 122 may expose the first insulating interlayer 102, and may be lower than the upper surface of the first insulating interlayer 102. The second portion of the bottom of each of the openings 122 may be at an inner portion of the first insulating interlayer 102.

[0087] In example embodiments, in a plan view, each of the openings 122 may have a line shape extending in one direction or a square shape.

[0088] In example embodiments, the upper surface of the first contact plug 112a exposed by each of the openings 122 may be hardly etched during the etching process. Accordingly, in each of the openings 122, the portion of the bottom exposing the first contact plug 112a may be higher than the other portion of the bottom exposing the first insulating interlayer 102.

[0089] In example embodiments, a lowermost surface of each of the openings 122 may be coplanar with the uppermost surface of the first barrier metal pattern 106b. In some example embodiments, the lowermost surface of each of the openings 122 may be lower than the uppermost surface of the first barrier metal pattern 106b.

[0090] In example embodiments, a lower width of the opening 122 may be less than a width of the first contact plug 112a. Accordingly, in the process for forming the openings 122, an entirety of the upper surface of the first contact plug 112a may not be exposed by each of the openings 122, but a portion of the upper surface of the first contact plug 112a may be exposed by each of the openings 122. Additionally, after forming the openings 122, at least a portion of the second insulating interlayer 120 filling the first recess 114 of the first contact plug 112a may remain without being removed.

[0091] Each of the openings 122 may have a sidewall slope such that a width at the bottom of the opening 122 is less than a width at an upper portion of the opening 122.

[0092] In example embodiments, a portion of the opening 122 positioned lower than the upper surface of the second insulating interlayer 120 may have a sidewall slope such that the width of the opening 122 gradually decreases toward the bottom thereof.

[0093] For example, as shown in FIG. 12, the opening 122 may have a sidewall slope such that the width gradually decreases from an upper portion to the bottom thereof. The sidewall slope may be constant. In this case, the interconnection structure shown in FIG. 1A or FIG. 1B may be formed by subsequent processes. For example, as shown in FIG. 13, a portion of the opening 122 positioned higher than the upper surface of the second insulating interlayer 120 has a vertical sidewall slope, and a portion of the opening 122 positioned lower than the upper surface of the second insulating interlayer 120 may have a sidewall slope such that the width of the opening 122 gradually decreases toward the bottom thereof. In this case, the interconnection structure shown in FIG. 1C may be formed by subsequent processes.

[0094] Referring to FIG. 14, a second barrier metal layer 130 may be conformally formed on sidewalls and bottoms of the openings 122 and on the upper surface of the second insulating interlayer 120. A second metal layer 132 may be formed on the second barrier metal layer 130 to fill the openings 122.

[0095] The second barrier metal layer 130 may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The second barrier metal layer 130 may be formed by a chemical vapor deposition process or an atomic layer deposition process.

[0096] In example embodiments, the second barrier metal layer 130 may be formed on the

sidewalls and on the bottoms of the openings **122** and the upper surface of the second insulating interlayer **120** to have a uniform thickness. In some example embodiments, a thickness of the second barrier metal layer **130** formed on the sidewalls of the openings **122** may be less than a thickness on the bottoms of the openings **122** and on the upper surface of the second insulating interlayer **120**.

[0097] In example embodiments, the second metal layer **132** may include a material having a resistance lower than a resistance of the first metal pattern **110a**. The second metal layer **132** may include, e.g., copper. In some example embodiments, the second metal layer **132** may include, e.g., aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0098] Referring to FIG. **15**, the second metal layer **124** and the second barrier metal layer **130** may be planarized until the upper surface of the second insulating interlayer **120** is exposed to form a first conductive pattern **134** filling each of the openings **122**. The first conductive pattern **134** may include a second barrier metal pattern **130a** and a second metal pattern **132a**. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0099] Since the first conductive pattern **134** is formed in each of the openings **122**, the first conductive pattern **134** may have a shape the same as a shape of an inner portion of each other the openings **122**.

[0100] Thereafter, a capping layer **140** may be formed on the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may cover upper surfaces of the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0101] FIG. **16** is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. **17** is an enlarged cross-sectional view of a portion of FIG. **16**.

[0102] The interconnection structure of the semiconductor device shown in FIG. **16** may be the same as the interconnection structure shown in FIG. **1A**, except that an adhesive layer is further included.

[0103] Referring to FIGS. **16** and **17**, an adhesive layer **116** may be on the first contact plug **112a** and the adhesive layer **116** may further be on the first insulating interlayer **102** to fill the first recess **114**. The adhesive layer **116** may cover upper surfaces of the first contact plug **112a** and the first insulating interlayer **102**. A second insulating interlayer **120** may be on the adhesive layer **116**.

[0104] The adhesive layer **116** may be disposed between the first insulating interlayer **102** and the second insulating interlayer **120** and the adhesive layer **116** may be further disposed between the first contact plug **112a** and the second insulating interlayer **120**.

[0105] The adhesive layer **116** may have a thickness less than a thickness of the second insulating interlayer **120**, and an upper surface of the adhesive layer **116** may be substantially flat. The adhesive layer **116** may include, e.g., a silicon oxide-based material. The adhesive layer **116** may be provided to improve adhesion between the first and second insulating interlayers **102** and **120**.

[0106] In example embodiments, a void may not be present in the adhesive layer **116** formed in the first recess **114**. In some example embodiments, a void may be present in the adhesive layer **116** formed in the first recess **114**.

[0107] The interconnection structure shown in FIG. **16** may be formed by performing the same processes as processes described with reference to FIGS. **6** to **15**, except that a process for forming the adhesive layer **116** may be further performed. First, the processes described with reference to FIGS. **6** to **10** may be performed. Thereafter, before forming the second insulating interlayer **120**, the adhesive layer **116** may be formed on the first insulating interlayer **102** and the adhesive layer **116** may be further formed on the first contact plug **112a** to fill the first recess **114**. Thereafter, the process described with reference to FIGS. **11** to **15** may be performed on the adhesive layer to form the interconnection structure shown in FIG. **16**.

[0108] FIG. **18** is a cross-sectional view of a semiconductor device according to example embodiments. FIG. **19** is a cross-sectional view of a semiconductor device according to some example embodiments.

[0109] The semiconductor device shown in FIGS. **18** and **19** may be a DRAM device.

[0110] Referring to FIG. **18**, a semiconductor device may be on a substrate **200** including a cell region A and a peripheral circuit region B.

[0111] The substrate **200** may be a wafer substrate including silicon, germanium, silicon-germanium, or a group III-V compound such as GaP, GaAs, GaSb, etc. In some example embodiments, the substrate **200** may be a Silicon On Insulator (SOI) wafer or a Germanium On Insulator (GOI) wafer.

[0112] Isolation layers **202** may be at the substrate **200**. The substrate **200** between the isolation layers **202** may be configured as an active region. A first active pattern **200a** may be in the cell region A, and a second active pattern **200b** may be in the peripheral circuit region B.

[0113] The cell region A of the substrate **200** may include a gate trench extending in a first direction parallel to an upper surface of the substrate **200**. A first gate structure may be in the gate trench. In example embodiments, the first gate structure may include a gate insulation layer, a gate electrode, and a capping pattern.

[0114] First impurity regions **205** configured as source/drain regions may be at upper portions of the first active patterns **200a** between the first gate structures. The first gate structure and first impurity regions **205** may be configured as a selection transistor. The selection transistor may be a recess channel transistor including the first gate structure buried in the substrate **200**.

[0115] An insulation layer pattern **206** may be on the first active pattern **200a**, the isolation layer **202**, and the first gate structure in the cell region A.

[0116] A bit line structure **220** may be on the insulation layer pattern **206** and the substrate **200**. The bit line structure **220** may be on the cell region A of the substrate **200**. The bit line structure **220** may contact one of the first impurity regions **205**.

[0117] The bit line structure **220** may include a first conductive pattern **220a**, a first barrier metal pattern **220b**, a first metal pattern **220c**, and a hard mask pattern **220d**. For example, the first conductive pattern **220a** may include polysilicon doped with impurities. The bit line structure **220** may extend in a second direction perpendicular to the first direction, and a plurality of bit line structures may be spaced apart from each other in the first direction. In example embodiments, a first spacer **222** may be on a sidewall of the bit line structure **220**.

[0118] Peripheral circuit patterns may be on the peripheral circuit region B of the substrate **200**. The peripheral circuit patterns may include, e.g., a peripheral circuit transistor **230**, lower wiring, etc. The peripheral circuit transistor **230** may include a second gate structure **224** and second impurity regions **228** configured as source/drain regions. The second impurity regions **228** may be at upper portions of the second active patterns **200b** adjacent to both sidewalls of the second gate structure **224**. The peripheral circuit transistor **230** may be configured as a sense amplifier, page buffer, command circuit, etc.

[0119] The second gate structure **224** may include a gate insulation layer pattern **207**, the first conductive pattern **220a**, the first barrier metal pattern **220b**, the first metal pattern **220c**, and the hard mask pattern **220d**. In example embodiments, an upper surface of the second gate structure **224** may be substantially coplanar with an upper surface of the bit line structure **220** in the cell region A. Each of the second gate structures **224** and the bit line structures **220** may include the first conductive pattern **220a**, the first barrier metal pattern **220b**, the first metal pattern **220c**, and the hard mask pattern **220d**. Therefore, the second gate structures **224** and the bit line structures **220** may have the same stacked structure.

[0120] In example embodiments, a second spacer **226** may be on a sidewall of the second gate structure **224**.

[0121] A first insulating interlayer **232** may occupy a space between the bit line structures **220**, and

may cover the sidewall of the second gate structure **224** on the peripheral circuit transistor **230**. In example embodiments, an upper surface of the first insulating interlayer **232** may be coplanar with upper surfaces of the bit line structure **220** and the second gate structure **224**.

[0122] A contact structure may pass through the first insulating interlayer **232** and the insulation layer pattern **206** on the cell region A, and the contact structure may contact a portion of the first impurity region **205**. The contact structure may have a structure in which a first lower contact plug **236** and a landing pad **244** are stacked.

[0123] The first lower contact plug **236** may be disposed between the bit line structures **220**, and a plurality of first lower contact plugs **236** may be spaced apart from each other. An upper surface of the first lower contact plug **236** may be lower than the upper surface of the bit line structure **220**. In example embodiments, the first lower contact plug **236** may include polysilicon.

[0124] The landing pad **244** may be on the first lower contact plug **236**. An upper surface of the landing pad **244** may be higher than the upper surface of the bit line structure **220**. A first upper insulating pattern **248a** may be disposed between the landing pads **244** in the horizontal direction. In example embodiments, an upper portion of the hard mask pattern **220d** in the bit line structure **220** may be partially etched through a patterning process, so that an upper portion of the hard mask pattern **220d** contacting the first upper insulating pattern **248** may have a recessed shape. In example embodiments, the landing pad **244** may include tungsten. In example embodiments, a barrier metal pattern may be further included at an interface between the landing pad **244** and the first lower contact plug **236**.

[0125] A second lower contact plug **242** may pass through the first insulating interlayer **232** on the peripheral circuit region B, and the second lower contact plug **242** may be connected to the second impurity regions **228** or the first metal pattern **220c** of the second gate structure **224**.

[0126] A second conductive pattern **246** may be on the first insulating interlayer **232** on the peripheral circuit region B. The second conductive patterns **246** may be wiring lines for electrically connecting the peripheral circuits. The second conductive patterns **246** may include the same material as the landing pad **244** on the cell region A.

[0127] A second upper insulating pattern **248b** may be between the second conductive patterns **246**. Upper surfaces of the second conductive patterns **246** and the second upper insulating pattern **248b** may be coplanar with upper surfaces of the landing pad **244** and the first upper insulating pattern **248a**.

[0128] A first etch stop layer **250** may be on the upper surfaces of the landing pad **244**, the second conductive patterns **246**, and the first and second upper insulating patterns **248a** and **248b**. The first etch stop layer **250** may include, e.g., silicon nitride.

[0129] A cell capacitor may pass through the first etch stop layer **250**, and may contact the landing pad **244**. The cell capacitor may include a plurality of lower electrodes **262**, a dielectric layer pattern **270a**, and an upper electrode **272a**. In addition, the cell capacitor may further include a lower support layer pattern **254a** and an upper support layer pattern **258a** for supporting the lower electrodes **262**.

[0130] The lower electrodes **262** may pass through the first etch stop layer **250**, and the lower electrodes **262** may be on the upper surface of each of the landing pads **244**, respectively. In example embodiments, the lower electrodes **262** may have a pillar shape. In some example embodiments, the lower electrodes **262** may have a cylindrical shape.

[0131] In example embodiments, the lower electrodes **262** may include a metal, e.g., Ti, W, Ni, or Co, or a metal nitride, e.g., TiN, TiSiN, TiAlN, TaN, TaSiN, or WN. For example, the lower electrodes **262** may include TiN.

[0132] Each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may be disposed between the lower electrodes **262**. Each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may be connected to sidewalls of the lower electrodes **262**, so that each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may support

the sidewalls of the lower electrodes **262**. The lower support layer pattern **254a** and the upper support layer pattern **258a** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0133] In some example embodiments, at least one of the upper support layer pattern **258a** and the lower support layer pattern **254a** may not be included. In some example embodiments, two or more first lower support layer patterns may be below the upper support layer pattern **258a**.

[0134] The dielectric layer pattern **270a** may conform to surfaces of the lower electrode **262**, the lower support layer pattern **254a**, the upper support layer pattern **258a**, and the first etch stop layer **250**. The dielectric layer pattern **270a** may include a high-k dielectric layer. In example embodiments, the high-k dielectric layer may include a metal layer, e.g., a hafnium oxide layer (HfO<sub>2</sub>), a zirconium oxide layer (ZrO<sub>2</sub>), an aluminum oxide layer (Al<sub>2</sub>O<sub>3</sub>), or a lanthanum oxide layer (La<sub>2</sub>O<sub>5</sub>).

[0135] The upper electrode **272a** may be on the dielectric layer pattern **270a**. In example embodiments, the upper electrode **272a** may include metal nitride. The metal nitride may include, e.g., titanium nitride (TiN), titanium silicon nitride (TiSiN), titanium aluminum nitride (TiAlN), tantalum nitride (Ta<sub>2</sub>N), tantalum silicon nitride (TaSiN), tantalum aluminum nitride (TaAlN), tungsten nitride (WN), or the like.

[0136] In example embodiments, the upper electrode **272a** may conform to the dielectric layer pattern **270a**, and may not completely occupy a space between the lower electrodes **262**.

[0137] The dielectric layer pattern **270a** and the upper electrode **272a** may be only on the cell region A of the substrate **200**. The dielectric layer pattern **270a** and the upper electrode **272a** may not be on the peripheral circuit region B of the substrate **200**. Accordingly, the dielectric layer pattern **270a** and the upper electrode **272a** B may be stacked on the upper surface of the first etch stop layer **250** at an edge portion of the cell region A adjacent to the peripheral circuit region B.

[0138] A first plate pattern **280** may be on the surface of the upper electrode **272a**. In example embodiments, the first plate pattern **280** may be only on the cell region A. The first plate pattern **280** may include, e.g., silicon germanium.

[0139] In example embodiments, the first plate pattern **280** may occupy a space between the upper electrodes **272a**, and the first plate pattern **280** may cover an uppermost surface of the upper electrode **272a**. Additionally, the first plate pattern **280** may be on the edge of the cell region A and on a sidewall of a last cell capacitor on the edge of the cell region A adjacent to the peripheral circuit region B. The first plate pattern **280** may not be on the peripheral circuit region B.

[0140] In example embodiments, the first plate pattern **280** may include a first portion, a second portion, and a third portion. The first portion may be positioned on the upper electrode **272a** on the cell region A, and an upper surface of the first portion may be substantially flat. The second portion may cover the last cell capacitor on the cell region A adjacent to the peripheral circuit region B, and may have a vertical surface perpendicular to the surface of the substrate **200**. The third portion may be connected to the second portion, and an upper surface of the third portion may be substantially flat. The third portion may be positioned on the edge of the cell region A adjacent to the peripheral circuit region.

[0141] An upper metal pattern **282** and a second etch stop pattern **284** may be stacked on the first plate pattern **280**.

[0142] A second insulating interlayer **286** may be on the second etch stop pattern **284** and the first etch stop layer **250**. An upper surface of the second insulating interlayer **286** may be substantially flat. The upper surface of the second insulating interlayer **286** may be higher than an upper surface of the second etch stop pattern **284**.

[0143] A first contact plug **302** may pass through the second insulating interlayer **286** and the first etch stop layer **250** on the peripheral region, and may contact each of the second conductive patterns **246**. The first contact plug **302** may be disposed in the first contact hole **292**.

[0144] A second contact plug **304** may pass through the second insulating interlayer **286** and the

second etch stop pattern **284** on the cell region, and may contact the upper metal pattern **282**. The second contact plug **304** may be disposed in the second contact hole **294** passing through the second insulating interlayer **286** and the second etch stop pattern **284** and exposing the upper metal pattern **282**.

[0145] The second contact plug **304** may be disposed over the cell capacitor. Accordingly, a bottom of the second contact plug **304** may be higher than a bottom of the first contact plug **302**. The upper surface of the second contact plug **304** and the upper surface of the first contact plug **302** may be coplanar with each other. A height in the vertical direction of the first contact plug **302** may be greater than a height in the vertical direction of the second contact plug **304**.

[0146] Each of the first and second contact plugs **302** and **304** may include a second barrier metal pattern **296a** and a second metal pattern **298**. The second barrier metal pattern **296a** may cover a portion of a sidewall of the second metal pattern **298** and a bottom of the second metal pattern **298**.

[0147] An upper surface of the second metal pattern **298** may be substantially coplanar with the upper surface of the second insulating interlayer **286**. Additionally, an uppermost surface of the second barrier metal pattern **296a** may be lower than the upper surface of the second metal pattern **298**. The second barrier metal pattern **296a** may not be on an upper sidewall of the second metal pattern **298**.

[0148] A first recess **300** may be disposed between the upper sidewall of the second metal pattern **298** and the second insulating interlayer **286**, and the first recess **300** may have a shape surrounding the upper sidewall of the second metal pattern **298**. The first recess **300** may be disposed above the uppermost surface of the second barrier metal pattern **296a**.

[0149] A third insulating interlayer **306** may be on the first contact plug **302**, the second contact plug **304**, and the second insulating interlayer **286**.

[0150] In example embodiments, the third insulating interlayer **306** may occupy the first recess **300**. A silicon oxide-based material may occupy the first recess **300**. The first recess **300** may not be occupied with nitride material, such as silicon nitride.

[0151] In example embodiments, the third insulating interlayer **306** in the first recess **300** may not include voids. In some example embodiments, the third insulating interlayer **306** in the first recess **300** may include voids. In some example embodiments, the third insulating interlayer **306** may not occupy the first recess **300**.

[0152] First and second openings **310** and **312** may pass through the third insulating interlayer **306**, and may extend into a portion under the upper surface of the second insulating interlayer **120** in the vertical direction. The first openings **310** may be disposed on the peripheral circuit region B, and the first contact plug **302** may be exposed by at least a portion of the first openings **310**. The second opening **312** are disposed on the cell region A, and the second contact plug **304** may be exposed by at least a portion of the second opening **312**.

[0153] Each of the first and second openings **310** and **312** may have a sidewall slope such that a lower width of each of the first and second openings **310** and **312** may be less than an upper width of each of the first and second openings **310** and **312**. A first conductive pattern structure **320** may be in each of the first openings **310**, and a second conductive pattern structure **324** may be in the second opening **312**.

[0154] A first bottom surface of the first conductive pattern structure **320** may contact an upper surface of the first contact plug **302**. A second bottom surface of the first conductive pattern structure **320** may contact the second insulating interlayer **286**. The third bottom surface of the second conductive pattern structure **324** may contact an upper surface of the second contact plug **304**. The fourth bottom surface of the second conductive pattern structure **324** may contact the second insulating interlayer **286**.

[0155] The first bottom surface of the first conductive pattern structure **320** may be positioned higher than the second bottom surface of the first conductive pattern structure **320**. The third bottom surface of the second conductive pattern structure **324** may be positioned higher than the

fourth bottom surface of the second conductive pattern structure **324**. The second and fourth bottom surfaces may be in the second insulating interlayer **286**. In example embodiments, the second and fourth bottom surfaces may be coplanar with or lower than a bottom of the first recess **300**.

[0156] Each of the first and second conductive pattern structures **320** and **324** may include a third barrier metal pattern **314** and a third metal pattern **316**. The third barrier metal pattern **314** may cover a sidewall and a bottom of the third metal pattern **316**.

[0157] The third metal pattern **316** may be on the third barrier metal pattern **314** to occupy the first and second openings **310** and **312**. Upper surfaces of the third metal pattern **316** and the third barrier metal pattern **314** may be substantially coplanar with an upper surface of the third insulating interlayer **306**.

[0158] In example embodiments, a width of an uppermost portion of each of the first and second conductive pattern structures **320** and **324** may be less than a width of a lowermost portion of the first and second conductive pattern structures **320** and **324**. In example embodiments, at least a lower portion of each of the first and second conductive pattern structures **320** and **324** may have a sidewall slope such that the width of the lower portion of each of the first and second conductive pattern structures **320** and **324** gradually decreases downward.

[0159] A portion of the first conductive pattern structure **320** lower than the upper surface of the second insulating interlayer **286** is referred to in the following disclosure as a fourth portion. In the claims, the recitation of a “fourth portion” without more is not intended to incorporate this definition therein. The fourth portion of the first conductive pattern structure **320** may be adjacent to the first contact plug **302** to be spaced apart from the first contact plug **302** in the horizontal direction.

[0160] In a cross-sectional view, the upper sidewall of the first contact plug **302** and the fourth portion of the first conductive pattern structure **320** may face toward each other in the horizontal direction.

[0161] A distance in the horizontal direction between an uppermost sidewall of the second metal pattern **298** of the first contact plug **302** and the first conductive pattern structure **320** may be greater than a distance in the horizontal direction between an uppermost sidewall of the second barrier metal pattern **296a** of the first contact plug **302** and the first conductive pattern structure **320**.

[0162] In example embodiments, an upper portion of the fourth portion of the first conductive pattern structure **320** may face the sidewall of the second metal pattern **298**, and a lower portion of the fourth portion of the first conductive pattern structure **320** may face the sidewall of the second barrier metal pattern **296a**. In some example embodiments, the fourth portion of the second conductive pattern structure **324** may face only the sidewall of the second metal pattern **298**.

[0163] The distance in the horizontal direction between the uppermost sidewall of the second barrier metal pattern **296a** of the first contact plug **302** and the first conductive pattern structure **320** is referred to in the following disclosure as a first distance. In the claims, the recitation of a “first distance” without more is not intended to incorporate this definition therein. Since the uppermost portion of the second barrier metal pattern **296a** has a recessed shape, the first distance may be greater than a first distance when an uppermost portion of a second barrier metal pattern is not recessed.

[0164] A region having the first distance may be a weak region where insulation breakdown of insulation material between the first contact plug **302** and the first conductive pattern structure **320** easily occurs, and thus reliability of the semiconductor device may be compromised. However, since an interface between the insulation materials is not positioned at the region having the first distance, reliability may be improved.

[0165] The second conductive pattern structure **324** may extend in one direction (e.g., longitudinal direction). In example embodiments, as shown in FIG. **18**, two or more second contact plugs **304**



arranged in a width direction perpendicular to the longitudinal direction may contact the bottom of the same second conductive pattern structure **324**. In some example embodiments, as shown in FIG. **19**, only one second contact plug **304** in the width direction may contact the bottom of the second conductive pattern structure **324**.

[0166] A capping layer **330** may be on the first conductive pattern structure **320**, the second conductive pattern structure **324**, and the third insulating interlayer **306**. The capping layer **330** may cover the first conductive pattern structure **320**, the second conductive pattern structure **324**, and the third insulating interlayer **306**. The capping layer **330** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0167] As described above, a portion of the first conductive pattern structure formed on the peripheral circuit region in the DRAM device may face the upper portion of the first contact plug disposed below the first conductive pattern structure in the horizontal direction. Since the upper portion of the barrier metal pattern included in the first conductive pattern structure has the recessed shape, a minimum distance between the first conductive pattern structure and the first contact plug may be increased. Additionally, the interface of the insulation material may not be positioned at a region having the minimum distance. Therefore, the DRAM device may have high reliability.

[0168] FIGS. **20** to **34** are cross-sectional views illustrating a method of manufacturing a semiconductor device according to example embodiments.

[0169] Referring to FIG. **20**, an isolation layer **202** for defining an active region may be formed on the substrate **200** including a cell region A and a peripheral circuit region B. Accordingly, a first active pattern **200a** may be formed on the cell region A, and the second active pattern **200b** may be formed on the peripheral circuit region B.

[0170] Select transistors may be formed on the cell region A of the substrate **200**. Each of the selection transistors may include a first gate structure and first impurity regions **205**. In example embodiments, a portion of the substrate **200** may be etched to form a gate trench, and the first gate structure may be formed in the gate trench. For example, the first gate structure may include a gate insulation layer, a gate electrode, and a capping pattern sequentially stacked.

[0171] An insulation layer pattern **206** may be formed on the cell region A of the substrate **200**. A recess may be formed on the first active pattern **200a** between the insulation layer patterns **206**. An upper surface of the first impurity region **205** may be exposed by a bottom of the recess. Additionally, a gate insulation layer may be formed on the peripheral circuit region B of the substrate **200**.

[0172] Bit line structures **220** may be formed on the insulation layer pattern **206** and the recess. The bit line structure **220** may include a first conductive pattern **220a**, a first barrier metal pattern **220b**, a first metal pattern **220c**, and a hard mask pattern **220d**. In example embodiments, a first spacer **222** may be formed on a sidewall of the bit line structure **220**.

[0173] When the bit line structure **220** is formed on the cell region A, a second gate structure **224** may be formed on the peripheral circuit region B of the substrate **200** together. The second gate structure **224** may include a gate insulation layer pattern **207**, a first conductive pattern **220a**, a first barrier metal pattern **220b**, a first metal pattern **220c**, and a hard mask pattern **220d** sequentially stacked. In example embodiments, a second spacer **226** may be formed on a sidewall of the second gate structure **224**. After forming the second gate structure **224**, second impurity regions **228** configured as source/drain regions may be formed at an upper portion of the second active pattern **200b** adjacent to both sidewalls of the second gate structure **224**.

[0174] Accordingly, a peripheral circuit transistor **230** including a second gate structure **224** and second impurity regions **228** may be formed on the peripheral circuit region B of the substrate **200**. The peripheral circuit transistor **230** may form a peripheral circuit.

[0175] A first insulating interlayer **232** may be formed on the substrate **200** to cover the bit line structures **220** and the peripheral circuit transistor **230**. Thereafter, an upper surface of the first

insulating interlayer **232** may be planarized. In example embodiments, the first insulating interlayer **232** may be planarized until upper surfaces of the bit line structure **220** and the second gate structure **224** are exposed.

[0176] A portion of the first insulating interlayer **232** between the bit line structures **220** may be etched to form a first lower contact hole **234** exposing a portion of the first impurity region **205** of the substrate **200**.

[0177] Referring to FIG. **21**, a first conductive layer may be formed on the first insulating interlayer **232** to fill the first lower contact hole **234**. Thereafter, the first conductive layer may be partially etched by an etch-back process to form a first lower contact plug **236** filling a lower portion of the first lower contact hole **234**. In the etch-back process, all of the first conductive layer formed on the first insulating interlayer **232** may be removed. In example embodiments, the first lower contact plug **236** may include polysilicon.

[0178] Thereafter, a portion of the first insulating interlayer **232** on the peripheral circuit region B may be etched to form a second lower contact hole **238** exposing the second impurity region **228** in the second active pattern **200b**.

[0179] Referring to FIG. **22**, a second conductive layer **240** may be formed to cover the bit line structure **220**, the first lower contact plug **236**, and the first insulating interlayer **232**. The second conductive layer may fill the first lower contact hole **234** and the second lower contact hole **238**. In example embodiments, the second conductive layer **240** may include tungsten. In some example embodiments, before forming the second conductive layer **240**, a process for forming a barrier metal layer may be further performed.

[0180] In example embodiments, before forming the second conductive layer **240**, a process for forming a metal silicide pattern on the first lower contact plug **236** may be further performed. In this case, the metal silicide pattern may also be formed on the surface of the substrate **200** exposed by a bottom of the second lower contact hole **238** on the peripheral circuit region B. By the above process, a second lower contact plug **242** may be formed in the second lower contact hole **238** on the peripheral circuit region B.

[0181] Referring to FIG. **23**, the second conductive layer **240** on the cell region A may be patterned to form a landing pad **244** contacting an upper surface of the first lower contact plug **236**. The second conductive layer **240** on the peripheral circuit region B may be patterned to form second conductive patterns **246**.

[0182] Patterning process for forming the landing pad **244** on the cell region A and the second conductive patterns **246** on the peripheral circuit region B may be performed by different photolithography processes or the same single photo lithography process.

[0183] In the patterning process, an upper portion of the hard mask pattern **220d** included in the bit line structure **220** may be partially etched, so that the upper portion of the bit line structure **220** may have a recessed shape.

[0184] Referring to FIG. **24**, an upper insulation layer may be formed on the landing pad **244**, the second conductive patterns **246**, and the first insulating interlayer **232**. The upper insulation layer may be planarized until upper surfaces of the landing pad **244** and the second conductive patterns **246** are exposed. The planarization process may include, e.g., a chemical mechanical polishing process.

[0185] Accordingly, a first upper insulating pattern **248a** may be formed in a gap between the landing pads **244**, and a second upper insulating pattern **248b** may be formed in a gap between the second conductive patterns **246**. Upper surfaces of the landing pad **244**, the second conductive pattern **246**, and the first and second upper insulating patterns **248a** and **248b** may be substantially coplanar with each other. Additionally, upper surfaces of the landing pad **244**, the second conductive pattern **246**, and the first and second upper insulating patterns **248a** and **148b** may be substantially flat.

[0186] A first etch stop layer **250** may be formed on the landing pad **244**, the second conductive

pattern **246**, and the first and second upper insulating patterns **248a** and **248b**. The first etch stop layer **250** may include, e.g., silicon nitride, silicon oxynitride, etc.

[0187] A first mold layer **252**, a lower support layer **254**, a second mold layer **256**, and an upper support layer **258** may be sequentially deposited on the first etch stop layer **250**.

[0188] The first and second mold layers **252** and **256** may include a material having an etch selectivity with respect to the lower support layer **254** and the upper support layer **258**. The first mold layer **252** and the second mold layer **256** may include, e.g., silicon oxide, and the lower support layer **254** and the upper support layer **258** may include, e.g., silicon nitride.

[0189] In this example embodiment, two support layers may be included in the semiconductor device, but may not be limited thereto. In some example embodiments, a single support layer or three or more support layers may be included in the semiconductor device, depending on a structure of a cell capacitor. In some example embodiments, the support layer may not be formed, and thus one mold layer may be included in the semiconductor.

[0190] An etch mask may be formed on the upper support layer **258**. The upper support layer **258**, the second mold layer **256**, the lower support layer **254**, and the first mold layer **252** and the first etch stop layer **250** may be etched using the etch mask to form first holes. The etching process may include an anisotropic etching process. By the above process, first holes **260** may expose the upper surface of the landing pad **244** on the cell region A of the substrate **200**.

[0191] Thereafter, a lower electrode layer may be formed to fill the first holes **260**. The lower electrode layer may be planarized until an upper surface of the upper support layer **258** is exposed to form lower electrodes **262** filling the first holes **260**. In this case, each of the lower electrodes **262** may have a pillar shape. The lower electrode **262** may include a metal, e.g., Ti, W, Ni, or Co, or a metal nitride such as TiN, TiSiN, TiAlN, TaN, TaSiN, or WN.

[0192] Referring to FIG. **25**, a first mask pattern may be formed on the upper support layer **258** and the lower electrode **262**. The first mask pattern may include amorphous carbon or polysilicon. The first mask pattern may include holes.

[0193] The upper support layer **258** may be anisotropically etched using the first mask pattern to form second holes exposing the upper portion of the second mold layer **256**. Accordingly, an upper support layer pattern **258a** may be formed on the cell region A of the substrate **200**. Thereafter, the second mold layer **256** may be wet etched by supplying etchant through the second holes. Accordingly, the lower support layer **254** may be exposed. Additionally, upper sidewalls of the lower electrodes **262** may be exposed.

[0194] A portion of the lower support layer **254** may be etched to form third holes exposing an upper portion of the first mold layer **252**. Accordingly, the lower support layer pattern **254a** may be formed. The first mold layer **252** may be wet etched by supplying etchant through the third holes.

[0195] The upper support layer pattern **258a** and the lower support layer pattern **254a** may be connected to each other while contacting a portion of an outer wall of the lower electrode **262**. Accordingly, sidewalls of the lower electrode **262** may be supported by the upper support layer pattern **258a** and the lower support layer pattern **254a**, so that a leaning of the lower electrode **262** may be prevented. Additionally, the surface of the lower electrode **262** may be exposed.

[0196] Thereafter, a dielectric layer **270** may be conformally formed along the surfaces of the lower electrode **262**, the lower and upper support layer patterns **254a** and **258a**, and the first etch stop layer **250**. An upper electrode layer **272** including metal may be formed on the dielectric layer **270**.

[0197] The dielectric layer **270** may include a high dielectric layer. In example embodiments, the high-k dielectric layer may include a hafnium oxide layer (HfO<sub>2</sub>), a zirconium oxide layer (ZrO<sub>2</sub>), an aluminum oxide layer (Al<sub>2</sub>O<sub>3</sub>), or a lanthanum oxide layer (La<sub>2</sub>O<sub>5</sub>).

[0198] In example embodiments, the upper electrode layer **272** may include a metal nitride.

[0199] Referring to FIG. **26**, a first plate layer may be formed on the upper electrode layer **272**. The first plate layer may include, e.g., a silicon germanium layer. The silicon germanium layer may be doped with, e.g., P-type impurities or N-type impurities.

[0200] In example embodiments, the silicon germanium layer may be formed by an atomic layer deposition process or a chemical vapor deposition process using a silicon source gas, a germanium source gas, and a dopant gas as deposition gases.

[0201] The silicon source gas may include, e.g., silane ( $\text{SiH}_4$ ) or dichloro silane ( $\text{SiH}_2\text{Cl}_2$ ). The germanium source gas may include, e.g., germanium ( $\text{GeH}_4$ ) or germanium tetrachloride ( $\text{GeCl}_4$ ). The dopant gas may include, e.g., borane ( $\text{BH}_3$ ), boron chloride ( $\text{BCl}_3$ ), phosphine ( $\text{PH}_3$ ), phosphorus chloride ( $\text{PCl}_3$ ), etc. Thereafter, the first plate layer may be crystallized by a heat treatment process.

[0202] Since the upper electrode layer **272** includes metal, it is not easy to increase a thickness of the upper electrode layer **272**. Therefore, if the upper electrode is formed only using the upper electrode layer **272**, a second contact plug subsequently formed may unintentionally pass through the upper electrode and extend into the lower electrode **262**. Therefore, the first plate layer may serve as a buffer layer for decreasing the likelihood of such a defect in subsequent processes.

[0203] The first plate layer may be conformally formed on the upper electrode layer **272** on the cell region A and the peripheral circuit region B to fill a gap between the upper electrode layers **272**.

[0204] An upper metal layer and a second etch stop layer may be formed on the first plate layer. The upper metal layer may include, e.g., tungsten. The second etch stop layer may include, e.g., silicon nitride or silicon oxynitride.

[0205] Thereafter, the second etch stop layer, the upper metal layer, the first plate layer, the upper electrode layer, and the dielectric layer on the peripheral circuit region B may be etched to form a dielectric layer pattern **270a**, an upper electrode **272a**, a first plate pattern **280**, an upper metal pattern **282**, and a second etch stop pattern **284**. A stacked structure of the dielectric layer pattern **270a**, the upper electrode **272a**, the first plate pattern **280**, the upper metal pattern **282**, and the second etch stop pattern **284** may be formed only on the cell region A.

[0206] The first plate pattern **280** may have a first portion having an upper surface higher than the upper surface of the lower electrode **262** on the cell region A, a second portion covering a last cell capacitor in the cell region A adjacent to the peripheral circuit region B, and a third portion connected to the second portion and disposed at the edge of the cell region. The second portion of the first plate pattern **280** may have a vertical surface perpendicular to the surface of the substrate **200**.

[0207] Referring to FIG. **27**, a second insulating interlayer **286** may be deposited on the first plate pattern **280** and the first etch stop layer **250**. The second insulating interlayer **286** may include a silicon oxide-based material. The second insulating interlayer **286** may include, e.g., a TEOS layer.

[0208] An upper surface of the second insulating interlayer **286** may be higher than an upper surface of the second etch stop pattern **284**. The upper surface of the second insulating interlayer **286** may be substantially flat. In example embodiments, a planarization process of the upper surface of the second insulating interlayer **286** may be further performed. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0209] Referring to FIG. **28**, the second insulating interlayer **286** and the first etch stop layer **250** on the peripheral circuit region B may be etched to form first contact holes exposing the second conductive pattern **246**. The second insulating interlayer **286** and the second etch stop pattern **284** on the cell region A may be etched to form second contact holes **294** exposing the upper metal pattern **282**.

[0210] In some example embodiments, example embodiments, the second contact hole **294** may have a sidewall slope such that a width of the second contact hole **294** may decrease from an upper portion of the second contact hole **294** to a bottom of the second contact hole **294**. The sidewall slope may be constant. Accordingly, the second contact hole **294** may have a lower width less than an upper width.

[0211] A second barrier metal layer may be formed on the surfaces of the first contact holes **292** and the second contact holes **294** and the second insulating interlayer **286**. A second metal layer

may be formed on the second barrier metal layer to fill the first contact holes **292** and the second contact holes **294**.

[0212] The second barrier metal layer may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The second metal layer may include, e.g., tungsten. In some example embodiments, the second metal layer may include, e.g., aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0213] The second metal layer and the second barrier metal layer may be planarized until the upper surface of the second insulating interlayer **286** is exposed to form a preliminary barrier metal pattern **296** and a second metal pattern **298a**. The planarization process may include, e.g., a chemical mechanical polishing process. The preliminary barrier metal pattern **296** and the second metal pattern **298a** may be formed in each of the first contact holes **292** and the second contact holes **294**.

[0214] Referring to FIG. **29**, an upper portion of the preliminary barrier metal pattern **296** may be partially etched to form a second barrier metal pattern **296a** in each of the first contact hole **292** and the second contact hole **294**.

[0215] Accordingly, a first contact plug **302** including the second barrier metal pattern **296a** and the second metal pattern **298** may be formed in the first contact hole **292**, and a second contact plug **304** including a second barrier metal pattern **296a** and a second metal pattern **298** may be formed in the second contact hole **294**.

[0216] In the first contact plug **302**, an uppermost surface of the second barrier metal pattern **296a** may be lower than an upper surface of the second metal pattern **298**. Additionally, in the second contact plug **304**, the uppermost surface of the second barrier metal pattern **296a** may be lower than the upper surface of the second metal pattern **298**. A first recess **300** may be formed between the second insulating interlayer **286** and the second metal pattern **298** above the upper surface of the second barrier metal pattern **296a**.

[0217] Referring to FIG. **30**, a third insulating interlayer **306** may be formed on the second insulating interlayer **286** and the first and second contact plugs **302** and **304** to fill the first recess **300**. The third insulating interlayer **306** may cover the second insulating interlayer **286** and the first and second contact plugs **302** and **304**. The same insulation material (e.g., the third insulating interlayer) may be formed in the first recess **300** and on the second insulating interlayer **286**, and the first and second contact plugs **302** and **304**.

[0218] Referring to FIGS. **31** and **32**, upper portions of the third insulating interlayer **306** and the second insulating interlayer **286** on the peripheral circuit region B may be etched to form a first opening **310** exposing at least a portion of the first contact plug **302**. A portion of the third insulating interlayer **306** and the second insulating interlayer **286** on the cell region A may be etched to form a second opening **312** exposing at least a portion of the second contact plug **304**.

[0219] The process for forming the first and second openings **310** and **312** may include a photo process and an etching process. The etching process may include, e.g., a dry etching process.

[0220] The first and second openings **310** and **312** may pass through the third insulating interlayer **306**, and extend into a portion under the upper surface of the second insulating interlayer **286** in the vertical direction.

[0221] In example embodiments, in a plan view, each of the first and second openings **310** and **312** may have a line shape or a square shape extending in one direction.

[0222] In example embodiments, as shown in FIG. **31**, a plurality of second contact plugs **304** arranged in the width direction may be exposed by a bottom of the second opening **312**. In some example embodiments, as shown in FIG. **32**, one second contact plug **304** in the width direction may be exposed by a bottom of the second opening **312**. Each of the first and second openings **310** and **312** may have a sidewall slope such that a lower width of each of the first and second openings **310** and **312** is less than an upper width of each of the first and second openings **310** and **312**.

[0223] In example embodiments, in each of the first and second openings **310** and **312**, at least a

portion lower than the upper surface of the second insulating interlayer **286** may have a sidewall slope such that a width is gradually decreased toward the bottom.

[0224] In the etching process for forming the first and second openings **310** and **312**, the first and second contact plugs **302** and **304** may be hardly etched. Accordingly, the bottom of the first opening **310** exposing each of the first contact plugs **302** may be higher than the bottom of the first opening **310** exposing the second insulating interlayer **286**, and the bottom of the second opening **312** exposing the second contact plugs **304** may be higher than the bottom of the second opening **312** exposing the second insulating interlayer **286**.

[0225] In example embodiments, at least one first opening **310** may expose only the second insulating interlayer **286** may be disposed between the two first openings **310** exposing first contact plugs **302**.

[0226] In example embodiments, at least one first opening **310** may be disposed between neighboring first contact plugs **302** in the horizontal direction. Only the second insulating interlayer **286** may be exposed by a lower sidewall of the first opening **310** disposed between the neighboring first contact plugs **302** in the horizontal direction.

[0227] Referring to FIGS. **33** and **34**, a third barrier metal layer may be conformally formed on sidewalls and on bottom surfaces of the first and second openings **310** and **312** and on the upper surface of the third insulating interlayer **306**. A third metal layer may be formed on the third barrier metal layer to fill the first and second openings **310** and **312**.

[0228] The third barrier metal layer may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. In example embodiments, the third metal layer may include a material having a resistance lower than a resistance of the second metal pattern **298**. For example, the third metal layer may include copper. In some example embodiments, the third metal layer may include aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0229] Thereafter, the third metal layer and the third barrier metal layer may be planarized until an upper surface of the third insulating interlayer **306** is exposed to form a third barrier metal pattern **314** and a third metal pattern **316**. The third barrier metal pattern **314** and the third metal pattern **316** may be formed in each of the first opening **210** and the second opening. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0230] In example embodiments, as shown in FIG. **33**, the second conductive pattern structure **324** may contact at least two second contact plugs **304** in the width direction. In this case, the semiconductor device shown in FIG. **18** may be manufactured by subsequent processes. In some example embodiments, as shown in FIG. **34**, the second conductive pattern structure **324** may contact only one second contact plug **304** arranged in the width direction. In this case, the semiconductor device shown in FIG. **19** may be manufactured by subsequent processes.

[0231] The first conductive pattern structure **320** may include a first bottom surface contacting a first contact plug **302** and a second bottom surface contacting the second insulating interlayer **286**. The second bottom surface of the first conductive pattern structure **320** may be lower than the first bottom surface of the first conductive pattern structure **320**.

[0232] A portion of the first conductive pattern structure **320** positioned lower than the upper surface of the second insulating interlayer **286** is referred to in the following disclosure as a fourth portion, and the fourth portion of the first conductive pattern structure **320** and an upper sidewall of the first contact plug **302** may face toward each other in the horizontal direction. In the claims, the recitation of a “fourth portion” without more is not intended to incorporate this definition therein.

[0233] In example embodiments, the sidewall of the second metal pattern **298** may be positioned higher than the uppermost surface of the second barrier metal pattern **296a** in the first contact plug **302**, and the sidewall of the second metal pattern **298** and the fourth portion of the conductive pattern structure **320** may face toward each other in the horizontal direction.

[0234] Referring to FIGS. **19** and **20** again, a capping layer **330** may be formed on the first and second conductive pattern structures **320** and **324** and the third insulating interlayer **306**. The

capping layer **330** may cover upper surfaces of the first and second conductive patterns **320** and **324** and the third insulating interlayer **306**. The capping layer **330** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0235] FIG. **35** is a cross-sectional view of a semiconductor device according to example embodiments.

[0236] The semiconductor device shown in FIG. **35** may be substantially the same as the semiconductor device shown in FIG. **19**, except that an adhesive layer is further included.

[0237] Referring to FIG. **35**, an adhesive layer **340** may occupy the first recess **300**, and the adhesive layer **340** may further cover upper surfaces of the first and second contact plugs **302** and **304** and the second insulating interlayer **286**. A third insulating interlayer **306** may be on the adhesive layer **340**.

[0238] In example embodiments, voids may not be present in the adhesive layer **340** in the first recess **300**. In some example embodiments, voids may be present in the adhesive layer **340** in the first recess **300**.

[0239] Processes for manufacturing the semiconductor device shown in FIG. **35** may be substantially the same as processes as those described with reference to FIGS. **20** to **34**, except for a process for forming the adhesive layer. First, the process described with reference to FIGS. **20** to **29** may be performed. Thereafter, before forming the third insulating interlayer **306**, the adhesive layer **340** may be formed on the second insulating interlayer **286** and the first and second contact plugs **302** and **304** to fill the first recess **300**. Thereafter, the processes described with reference to FIGS. **30** to **34** may be performed on the adhesive layer **340** to manufacture the semiconductor device.

[0240] The interconnection structure included in the semiconductor device may have high reliability, so the semiconductor device may have high reliability.

[0241] While the present inventive concepts have been shown and described with reference to example embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the scope of the present inventive concepts as set forth by the following claims.

## Claims

1. A semiconductor device, comprising: a substrate; a first insulating interlayer on the substrate; a plurality of first contact plugs passing through the first insulating interlayer, each of the plurality of first contact plugs including: a first metal pattern; and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern; a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction, each of the plurality of first conductive patterns having: a first bottom surface contacting a portion of an upper surface of a respective first contact plug of the plurality of first contact plugs; and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.
2. The semiconductor device of claim 1, wherein an upper surface of the first insulating interlayer and an upper surface of the first metal pattern are coplanar with each other, and a first recess is between the first insulating interlayer and the first metal pattern.
3. The semiconductor device of claim 2, wherein the second insulating interlayer occupies the first recess.
4. The semiconductor device of claim 2, further comprising an adhesive layer that is between the first insulating interlayer and the second insulating interlayer and is further between the plurality of

first contact plugs and the second insulating interlayer, and wherein the adhesive layer occupies the first recess.

**5.** The semiconductor device of claim 1, wherein an uppermost surface of the first barrier metal pattern is lower than an upper surface of the first metal pattern, and wherein each of the plurality of first conductive patterns contacts each of: the upper surface of the first metal pattern; the upper sidewall of the first metal pattern; and an upper surface of the first barrier metal pattern.

**6.** The semiconductor device of claim 1, wherein the second bottom surface of each first conductive pattern is coplanar with or lower than an upper surface of the first barrier metal pattern of the respective first contact plug.

**7.** The semiconductor device of claim 1, wherein at least a portion of each first conductive pattern is lower than the upper surface of the first insulating interlayer and faces the upper sidewall of the first metal pattern of the respective first contact plug in a horizontal direction.

**8.** The semiconductor device of claim 1, wherein each of the first conductive patterns has a sidewall slope such that a lower width of the first conductive pattern is less than an upper width of the first conductive pattern.

**9.** The semiconductor device of claim 1, wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the respective first contact plug and the first conductive pattern is greater than a distance in the horizontal direction between the uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

**10.** The semiconductor device of claim 1, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first insulating interlayer.

**11.** The semiconductor device of claim 1, further comprising a capping layer covering the plurality of first conductive patterns and further covering the second insulating interlayer.

**12.** A semiconductor device, comprising: a substrate; a first insulating interlayer on the substrate, the first insulating interlayer including a plurality of contact holes; a plurality of first contact plugs in the plurality of contact holes, respectively, each first contact plug including: a first metal pattern; and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and an uppermost surface of the first barrier metal pattern being lower than an upper surface of the first metal pattern; a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction, and each of the plurality of first conductive patterns contacting an upper surface of a respective first contact plug of the plurality of first contact plugs and further contacting a respective upper portion of the first insulating interlayer, wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the respective first contact plug and the first conductive pattern is greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

**13.** The semiconductor device of claim 12, wherein at least a portion of each first conductive pattern is lower than the upper surface of the first insulating interlayer and faces the first metal pattern of the respective first contact plug in a horizontal direction.

**14.** The semiconductor device of claim 12, wherein the upper surface of the first insulating interlayer and the upper surface of the first metal pattern are coplanar with each other, and a first recess is between the first insulating interlayer and the first metal pattern.

**15.** The semiconductor device of claim 14, wherein the second insulating interlayer occupies the first recess.

**16.** The semiconductor device of claim 12, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first



insulating interlayer.

**17.** The semiconductor device of claim 12, wherein each of the plurality of first conductive patterns includes a first bottom surface contacting a portion of an upper surface of the respective first contact plug, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

**18.** A semiconductor device, comprising: a substrate; a plurality of capacitors on a cell region of the substrate; a plurality of peripheral circuit patterns on a peripheral circuit region of the substrate; a first insulating interlayer covering the plurality of capacitors and further covering the plurality of peripheral circuit patterns; a plurality of first contact plugs passing through the first insulating interlayer and connecting to the plurality of peripheral circuit patterns, each of the plurality of first contact plugs including: a first metal pattern; and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern; a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and a first conductive pattern passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction, and the first conductive pattern contacting upper surfaces of the plurality of first contact plugs and further contacting an upper surface of the first insulating interlayer on the peripheral circuit region, wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of a first contact plug of the plurality of first contact plugs and the first conductive pattern is greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the first contact plug and the first conductive pattern.

**19.** The semiconductor device of claim 18, further comprising: a plurality of second contact plugs passing through the first insulating interlayer and electrically connecting to respective capacitors of the plurality of capacitors, each of the plurality of second contact plugs including: a first metal pattern; and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern; and second conductive patterns contacting an upper surface of each of the plurality of second contact plugs and contacting an upper surface of the first insulating interlayer on the cell region.

**20.** The semiconductor device of claim 18, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first insulating interlayer.

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